Number Bonds to 10 — Fixing an Incomplete or Duplicated Family
Answer Key

Kindergarten–Grade 1

Quick Answers

1. 2	3. 7	5. 3
2. 6	4. 4	6. 3, 5

Common wrong answers

2. *If they got 12: added the whole and the part instead of taking the part away*
3. *If they got 2: wrote the take-away fact 9 minus 7 a second time instead of the missing one*
4. *If they got 10: wrote the adding fact 6 plus 4 again instead of the missing take-away fact*

1. Warm up: the bond shows the whole 6 and one part 4. Write the missing part.

Step 1: name what the bond shows — the top circle is the whole (6) and the two lower circles are the parts, so one part plus 4 must make 6. Step 2: count on from 4 up to 6: 4, then 5, then 6 — that is 2 more. Step 3: check by adding the parts back together: $4 + 2 = 6$ ✓. The missing part is $\boxed{2}$.

2. Mia writes 12 for the missing part of the bond with whole 9 and part 3.

Step 1: name Mia's mistake — she put the two numbers she saw together ($9 + 3 = 12$). Adding gives a number *bigger* than the whole, and a part can never be bigger than its whole, so 12 cannot be right. Step 2: a missing *part* is found by taking the known part away from the whole: $9 - 3$. Step 3: take 3 away from 9: what is left is 6. Step 4: check: $3 + 6 = 9$ ✓. The missing part is $\boxed{6}$.

3. Ben's family for parts 2 and 7 (whole 9): $2+7=9$, $7+2=9$, $9-7=2$, $9-7=2$. Cross out the repeat and write the missing fact.

Step 1: sort Ben's four facts — two adding facts ($2 + 7 = 9$ and $7 + 2 = 9$) and *the same* take-away fact written twice ($9 - 7 = 2$). Step 2: every family has two take-away facts, one for each part. Ben took away the part 7; he never took away the part 2. Step 3: cross out the second $9 - 7 = 2$ and take the other part away instead: $9 - 2$ leaves 7. Step 4: check that the answer is the other part of the bond: yes, 7 is a part ✓. The missing fact is $9 - 2 = \boxed{7}$.

4. Ana's family for parts 6 and 4 (whole 10): $6 + 4 = 10$, $4 + 6 = 10$, $10 - 4 = 6$, $6 + 4 = 10$. Cross out the repeat and write the missing fact.

Step 1: count the kinds of fact — Ana wrote *three* adding facts (the third one, $6 + 4 = 10$, repeats the first) and only *one* take-away fact. Step 2: a family is two adding facts and two take-away facts, so what is missing is a take-away fact — the one that takes away the other part, 6. Step 3: cross out the repeated $6 + 4 = 10$ and compute $10 - 6$, which leaves 4. Step 4: check: $6 + 4 = 10$ ✓, so $10 - 6 = \boxed{4}$.

5. Jo's family for parts 7 and 3 (whole 10): the last fact is $10 - \_\_\_\_ = 7$. Write the missing number.

Step 1: read the fact — something is taken away from the whole 10 and the answer left over is the part 7. Step 2: in a bond, taking one part away leaves the *other* part. Since 7 is left, the number taken away must be the other part of the bond. Step 3: the parts are 7 and 3, so the number taken away is 3. Step 4: check the finished fact: $10 - 3 = 7$ ✓. The missing number is $\boxed{3}$.

6. Challenge: Sam's bond has parts 5 and 3 (whole 8) and only the two adding facts. Write both missing take-away facts.

Step 1: list what Sam has — $5 + 3 = 8$ and $3 + 5 = 8$. Those are both *adding* facts, so both *take-away* facts are missing. Step 2: a bond makes one take-away fact for each part. Take away the first part: $8 - 5$ leaves the other part. Step 3: $8 - 5 = \boxed{3}$. Step 4: now take away the other part: $8 - 3$ leaves 5, so $8 - 3 = \boxed{5}$. Step 5: check both against the bond — each answer is the part Sam did not take away ✓.